Agentic Knowledge Graphs for Research Progression

PhD Qualifying Exam Presentation

Jason Cusati

Department of Computer Science
Virginia Tech
djjay@vt.edu

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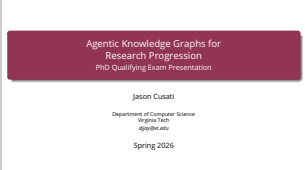

Good morning. I'm Jason Cusati, a PhD student working with Dr. Brown at the Code World No Blanket Lab here at Virginia Tech. The lab name is fitting, actually — today I'm arguing that AI research systems need somewhere to put their knowledge, because right now they're working without a net. Or a blanket. Today's talk is Agentic Knowledge Graphs for Research Progression — the academic way of saying: what if AI actually remembered what it learned, and could connect the dots — not just within a single paper, but across a whole field of research?

Outline

1. Motivation
2. Literature Survey
3. Synthesis
4. Proposal
5. Evaluation
6. Contributions
7. Conclusion

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Here's the plan for the next 15 minutes. I'll start with the problem — why current AI-assisted workflows keep starting from scratch. Then I'll survey five domains and show where the gaps are. From there I'll present the proposal: a provenance-first architecture that gives agents a persistent, structured memory. I'll close with the evaluation framework and contributions. Questions at the end — I'm told there will be some, so I've been practicing my nervous smile.

The Structural Problem in AI-Assisted Research

- Research reasoning is **fragmented** and **transient**
- LLM-assisted workflows **amplify volatility**
- **Structural memory** is missing

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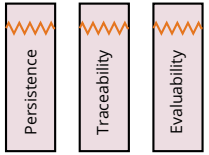
Current AI systems dramatically accelerate document generation and research exploration.
However, they do not accumulate structured, validated, provenance-aware knowledge across iterations.
Outputs are transient. Each interaction effectively starts from scratch.
This work investigates how agentic systems combined with knowledge graphs can provide a persistent substrate for research progression.

The Structural Gaps

- **Persistence:** State beyond session boundaries
- **Traceability:** Linkage from Claim → Evidence
- **Evaluability:** Empirical measurement of progression



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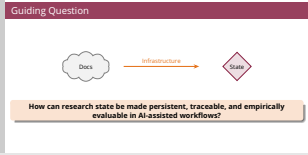
This manifests as three specific structural gaps.
First, Persistence: We lack state that survives beyond a single chat session. Second, Traceability: When an agent makes a claim, we often cannot trace it back to the specific evidence that supported it. Third, Evaluability: We lack the tools to empirically measure if our research knowledge is actually progressing over time.
Crucially, these are infrastructure problems, not model quality problems. Better models won't fix this — better architecture will.

Guiding Question

How can research state be made persistent, traceable, and empirically evaluable in AI-assisted workflows?



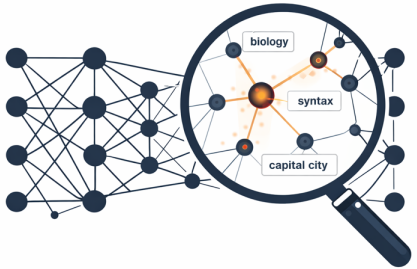
This leads to the guiding question of my proposal:
How can we transform this ephemeral interaction into something solid? How can research state be made persistent, traceable, and empirically evaluable in AI-assisted workflows?
To answer this, I've surveyed the current landscape to understand what pieces we already have — and what is missing.

Landscape Overview

I surveyed five key domains that contribute to this problem.
We look at Mechanistic Interpretability to understand how models represent knowledge internally. We look at RAG and Knowledge Graphs as two ways of representing knowledge externally. We look at Agentic Systems which act on that knowledge. And finally, Empirical Standards to understand how we should evaluate these systems.

Mechanistic Interpretability

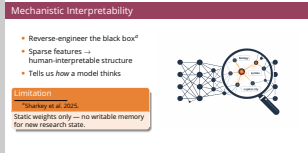
- Reverse-engineer the black box[a]
- Sparse features → human-interpretable structure
- Tells us *how* a model thinks



Limitation

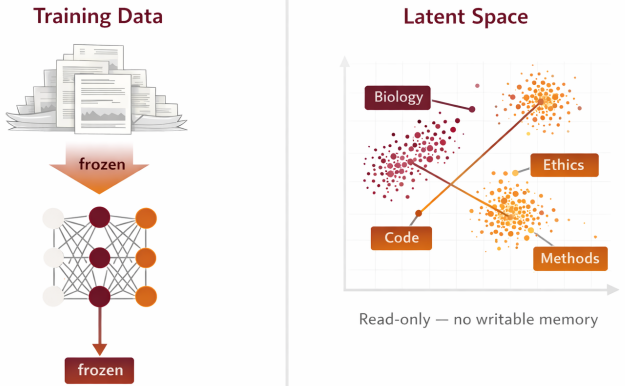
[a] Sharkey et al. 2025.

Static weights only — no writable memory for new research state.



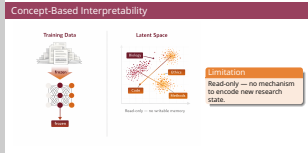
Starting with Mechanistic Interpretability, work by Sharkey et al. focuses on reverse-engineering the black box.
Techniques like sparse dictionary learning allow us to decompose model weights into understandable features. This is critical for understanding how a model thinks.
However, the limitation is that this focuses on the model's internal, static weights. It does not provide a mechanism for storing new research state that an agent generates during a workflow.

Concept-Based Interpretability



Limitation

Read-only — no mechanism to encode new research state.

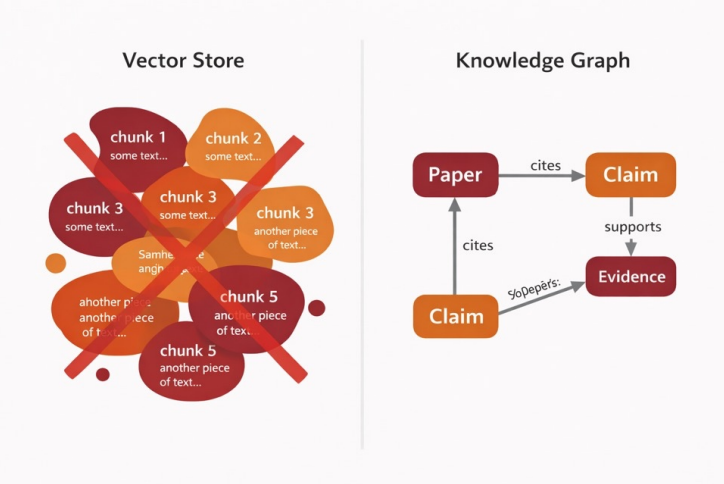


Concept-based interpretability takes a different angle. Instead of looking at circuits and weights, it asks: can we find directions in the model's latent space that align with human-readable concepts — like "Biology" or "Code"?

And the answer is kind of yes. Probes can find those directions. But what we're really measuring is what the model learned from its training data. It's sophisticated pattern matching on frozen knowledge.

The problem is that probing is fundamentally read-only. It tells us what the model baked in during training — not what we discovered in this session. Close the tab, and it's gone.

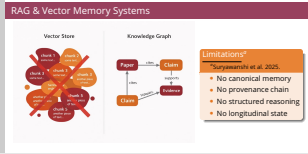
RAG & Vector Memory Systems



Limitations[a]

[a] Suryawanshi et al. 2025.

- No canonical memory
- No provenance chain
- No structured reasoning
- No longitudinal state

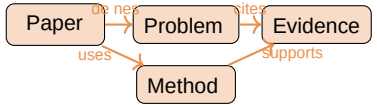


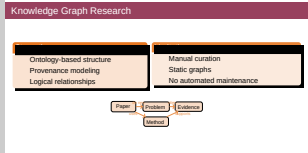
RAG is the go-to for external memory right now, and for good reason — it scales, it's fast, and semantic similarity gets you surprisingly far.
But Suryawanshi et al. point out the ceiling: vector retrieval gives you chunks, not structure. Similarity, not reasoning.
And critically, there's no canonical memory. Ask the same question after new papers drop and you get a different answer. No provenance chain — you can't trace a generated claim back to the specific text that justified it.

Knowledge Graph Research

Ontology-based structure
Provenance modeling
Logical relationships

Manual curation
Static graphs
No automated maintenance





Knowledge Graphs o er exactly the structure that RAG lacks | explicit ontologies, logical relationships, and provenance modeling baked in. Suryawanshi et al. show how this enables cross-document reasoning that vector retrieval simply can't do.
But traditional KGs are high-friction. They require domain experts to curate them manually, they're static snapshots, and they don't evolve as the literature grows. A graph built today is already stale tomorrow.
The gap is automation. We have structure from KGs and we have automation from LLMs | but they're rarely integrated in a way that lets the graph actually evolve. That integration is exactly what this proposal targets.

Empirical Evaluation Standards

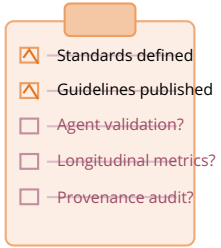
- Ralph et al. (2021): Empirical Standards for SE
- Baltes et al. (2025): Guidelines for LLM Studies

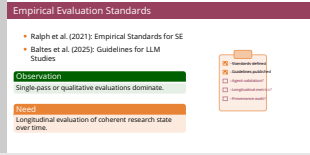
Observation

Single-pass or qualitative evaluations dominate.

Need

Longitudinal evaluation of coherent research state over time.



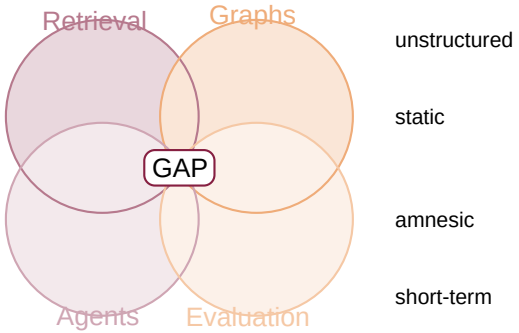


Finally, how do we evaluate any of this? Ralph et al. and Baltes et al. give us strong standards for SE and LLM experiments — clear, rigorous, and well-respected in the community.

But current agentic research frequently falls short. Evaluations are single-pass or qualitative. We test a snapshot, not a trajectory.

What's missing is longitudinal evaluation — frameworks that measure whether an agent is actually maintaining a coherent research state over weeks or months, not just getting the right answer once.

Cross-Cutting Gaps in the Literature



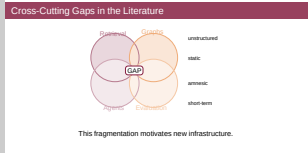
unstructured

static

amnesic

short-term

This fragmentation motivates new infrastructure.



Synthesizing across all ve domains, the fragmentation is clear.
Retrieval is unstructured. Graphs are static. Agents are amnesic. Evaluation is short-term. Each domain has made real progress in isolation | but none of them connect.
Retrieval nds content but can't reason over it. Graphs provide structure but require human maintenance. Agents can execute complex tasks but forget everything when the session ends. And our evaluation tools weren't designed for longitudinal, multi-session work ows.

No existing system integrates persistent structured memory with autonomous agency and rigorous provenance. That gap in the center isn't an accident | it's the core blocker to true AI-assisted research progression.

Contributions

1. **Formal definition** of Research State in an AI context
2. **Agentic Curation Protocol** for graph maintenance
3. **Reference infrastructure** implementation

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Near-Term Work

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- Artifact release (Code & Datasets)
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Conclusion

Fragmented Chat → **Persistent State**

- **From:** Fragmented chat interaction
- **To:** Persistent, traceable research state
- **Goal:** Infrastructure for discovery

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- From: Fragmented chat interaction
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Questions?

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